Pretrained VS Custom CNN Diagnosis

Task: Car body type classification
Models Compared: Custom CNN, Resnet18 with un/frozen backbone
Dataset: Car Body Types Dataset (7550 images), 70/15/15 split.
Hyperparameters: Padding, Kernel Size, Data Augmentation, ReduceLROnPlateau, LR, Batch size, Weight Decay, Dropout Probability, Optimizer Function, Number of Epochs

Parameter	**Custom CNN**	**Resnet18_Frozen**	**Resnet18**
Backbone	4 Conv. Blocks	Frozen	Unfrozen Layer 4
Class. Head	Linear (128->7)	Linear(512->7)	Linear(512->7)
N Parameters	~400k	~3.5k	~11.8m
Learning Rate	Constant (1e-4)	Constant (1e-4)	Layer 4: 1e-4, FC: 1e-3
Epochs	100	10	10

Custom CNN Settings

Sequential CNN Layers: (3 -> 32 -> 64 -> 128 -> 256 features),
Head: (256 -> 128)
ReduceLROnPlateau: patience - 8, mode - min, scale - 0.2
Filter size: 3x3,
Epochs Count: 100,
Lr: 1e-4,
Batch size: 64,
Weight Decay: 1e-3,
Dropout Probability: 0.1,
Optimizer: AdamW

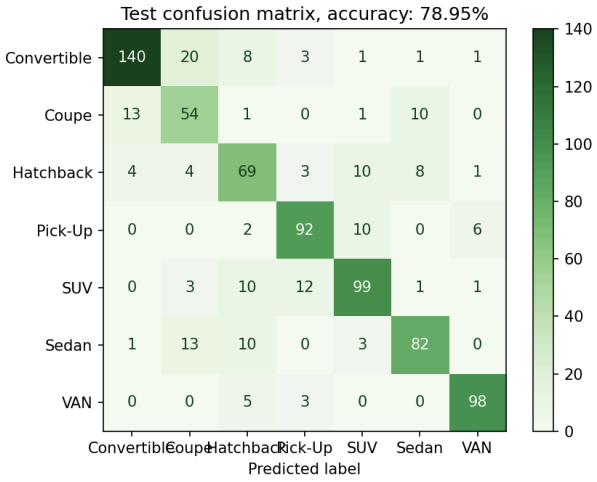
Performance Comparison

Parameter	Custom CNN	Resnet18_Frozen	Resnet18
Test Accuracy	79%	79%	92%
Training Loss	0.53	0.56	0.17
Training time	2960 sec.	290 sec.	304 sec.
Training time Per Epoch	29.6 sec.	29 sec.	30 sec.
Inference time	6.5 ms	6.97 ms	7.35 ms
Customizability	High	Low	Medium
Available Param.	~400k	~3.5k	~11.8m

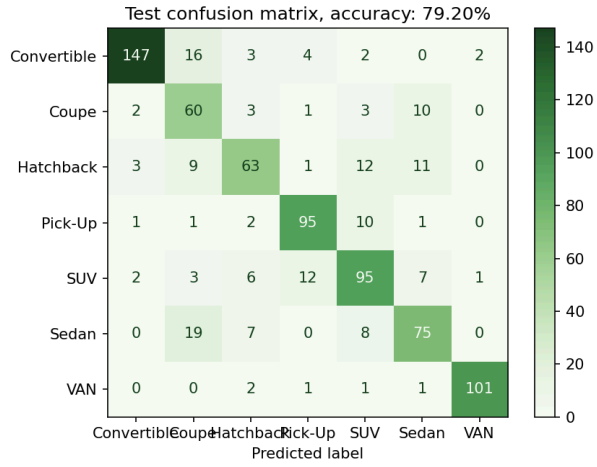
Resnet18 has a big advantage due to it being pretrained on millions of images from Imagenet. Its first layers already know how to classify border lines, curves and general features, the only thing that's left is to adapt to specific objects from the dataset. We can see that in action when the outer layer of backbone is unfrozen: model yields high accuracy and low loss values. The version with frozen backbone is very close to Custom CNN in accuracy, but has slightly higher loss values. That is explained by the fact that custom model was learning features specific to the dataset from ground-up, whilst the frozen resnet18 had limitation of only adjusting head dense layer.
Adding data augmentation to Custom model has improved accuracy and loss values, still model training accuracy flatlines when approaching 80%.
Possible improvement would be K-fold Cross Validation, since the dataset isn't large, rotating the validation set values periodically would result in better learning speed and accuracy. However, for this particular task this technique wasn't used to better observe the impact of Data Augmentation.

Confusion Matricies

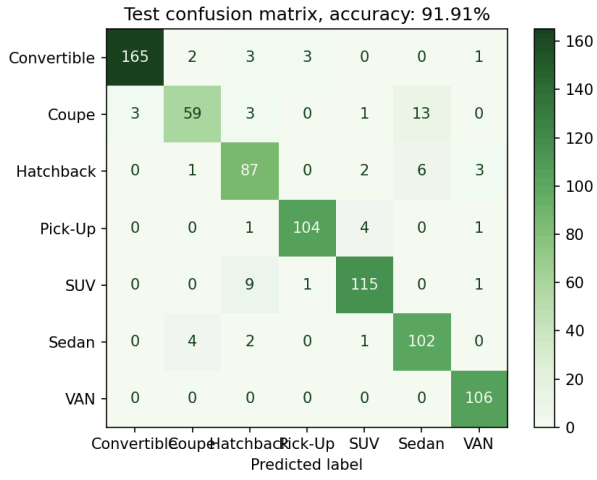
The most commonly confused class is Coupe, hence its similarity to Sedan or Hatchback



Custom CNN



Resnet18_Frozen

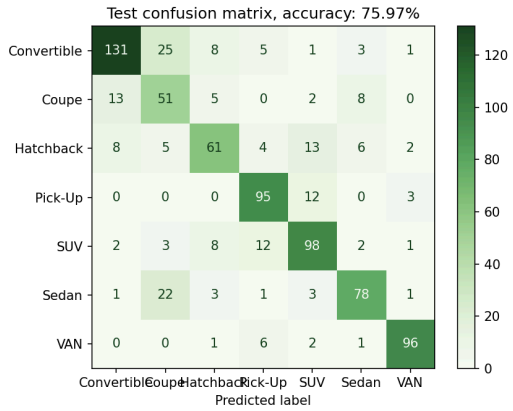


Resnet18

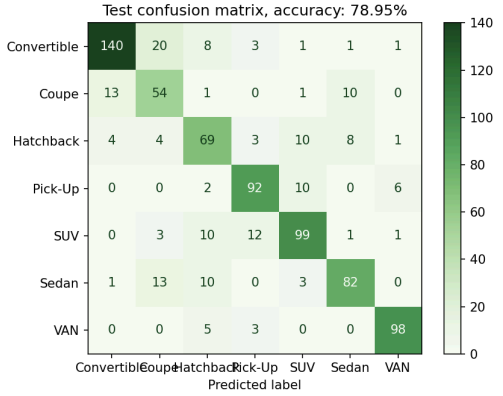
Verdict

Custom CNN Use Cases:
The task is highly specific, so that model pretrained on generic dataset would be of no use (non-natural imagery).
When a model requires deep and more complex, or contrary, simple architecture for specific hardware. When full control over model architecture is desired.
A good choice when the goal is to learn how CNN architecture works behind the scenes.
Frozen Backbone Use Cases:
Hardware capacity is severely limited and/or the dataset is extremely small.
Can adapt quickly to new datasets, with compromise on accuracy.
Since the dense layer is the only one being trained and also is decision making, the number of trainable parameters is limited.
Partial Fine-Tuning Use Cases:
For non-specific visual classification tasks.
Model can pick up visual clues and patterns on low levels, retrained high levels produce accurate predictions for imagery related to one the low layers were trained on.
Works well for where high-level features need domain adaptation.
Relatively cheap and fast training process resulting in high accuracy.
Limited control over model architecture.

Data Augmentation Impact



Data Augmentation: Off



Data Augmentation: On

Adding random image rotation, horizontal flips and jittering has improved accuracy dramatically for Custom CNN model, since it adds variation to the dataset and allows model to learn specific features more precisely, resulting in better accuracy and lower loss. Accuracy improved from 75.9% to 78.9%, loss went from 0.7 to 0.53 respectively.